

SE – Overview of IT Industry

Name : Panchal Heer

Subject : Software Engineering-Overview of IT Industry

Session 3 : Internet and Networking Basics

A computer network is a system of interconnected devices that communicate to share data and resources.

- The internet is simplified the “network of networks”-a massive global infrastructure linking billions of devices using shared communication rules (protocols) and physical connections.

Task 1 : Full Form and Uses

Term	Full Form	Use
HTTP	HyperText Transfer Protocol	Transfers web pages over the internet
HTTPS	HyperText Transfer Protocol Secure	Secure communication on websites
FTP	File Transfer Protocol	Transfer files between computers
SMTP	Simple Mail Transfer Protocol	Sends emails over the internet

Task 2 : IPv4 vs IPv6

- IPv4 uses 32-bit addresses and supports fewer unique addresses.
- IPv6 uses 128-bit addresses and supports a much larger number of addresses.

Task 3 : nslookup google.com

If you can't run nslookup , write :

nslookup google.com

Result :

Multiple IP addresses may be returned depending on location and network.

Task 4 : HTTP Status Codes

Code	Meaning
200	OK (Request Successful)
301	Moved Permanently
403	Forbidden
404	Not Found
500	Internal Server Error

Task 5 : DNS Hierarchy Example

Using www.google.com

- Root = .
- TLD = .com

- Domain = google.com
- Subdomain = www

Conclusion :

This session covers IT career roles , software architecture , client-server communication , databases , internet fundamentals , networking concepts , and DNS hierarchy.